

Exercises: Constrained Optimization

Exercise 1 (Penalization). We consider the following problem

$$(P) \quad \min_{x \in \mathbb{R}^n} \quad f(x) \\ \text{s.t.} \quad Ax = b, \quad x \leq 0$$

with value v and the following penalized versions

$$(P_t^{in}) \quad \min_{x \in \mathbb{R}^n} f(x) - t \sum_{i=1}^n \ln(-x_i) \\ \text{s.t.} \quad Ax = b, \quad x < 0$$

and

$$(P_t^{out}) \quad \min_{x \in \mathbb{R}^n} f(x) + t \sum_{i=1}^n (x_i)^+ \\ \text{s.t.} \quad Ax = b$$

with associated value v_t^{in} and v_t^{out} , and an optimal solution x_t^{in} and x_t^{out} .

1. Intuitively, assuming that f is "well behaved", for t going to which value does (P_t^{in}) tend to the original problem (P) ? In which sense?
2. What can you say about x_t^{in} ?
3. Can you compare v_t^{in} and v ?
4. Same questions for (P_t^{out}) .

Exercise 2 (A naive feasible direction). Let $X = \mathbb{R}_+^2$ and $f(x_1, x_2) = (x_1 - 1)^2 + (x_2 - 1)^2$. Consider the following "masked gradient" direction:

$$d_i(x) = -\nabla_i f(x) \mathbb{1}_{x_i > 0}.$$

1. Is $d(x)$ a feasible descent direction?
2. Apply the corresponding algorithm from $x^{(0)} = (0, 1)$ and comment the result.

Exercise 3 (Decomposition by prices). We consider the following energy problem:

- you are an energy producer with N production units
- you have to satisfy a given demand profile for the next 24h (i.e. the total output at time t should be equal to d_t)
- the time step is the hour, and each unit has a production cost over the horizon given as a convex quadratic function of the production profile
- For each unit i , the production planning $u^i = (u_t^i)_{t \in [24]}$ has to satisfy polyhedral constraints $u^i \in U^i$.

1. Model this problem as an optimization problem. To which class does it belong? How many variables?
2. Apply Uzawa's algorithm to this problem. Why could this be an interesting idea?
3. Give an economic interpretation to this method.
4. What would happen if the production units had coupling constraints between them? For example, if at any time step the sum of the production of units 1 and 2 could not exceed a given amount M_t .
5. What would happen if each unit had start-up costs?

Exercise 4 (Kelley's convergence). We are going to prove that, if $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is convex, and X a non-empty polytope (bounded polyhedron), then Kelley's cutting plane algorithm converges.

Consider $x^{(1)} \in X$. Given $x^{(1)}, \dots, x^{(k)}$, with $g^{(\kappa)} \in \partial f(x^{(\kappa)})$ for all $\kappa \in [k]$, define $(x^{(k+1)}, \underline{v}^{(k+1)})$ as an optimal solution to $\mathcal{P}^{(k)}$:

$$\begin{aligned} \underline{v}^{(k+1)} = \operatorname{Min}_{x \in X, z \in \mathbb{R}} \quad & z \\ \text{s.t.} \quad & f(x^{(\kappa)}) + \langle g^{(\kappa)}, x - x^{(\kappa)} \rangle \leq z, \\ & \kappa \in [k]. \end{aligned}$$

and then choose $g^{(k+1)} \in \partial f(x^{(k+1)})$.

Denote $v = \min_{x \in X} f(x)$.

1. Show that v exists and is finite, and that there exists such a sequence $(x^{(k)})_{k \in \mathbb{N}^*}$.
2. Show that there exists L such that, for all k_1 and k_2 , we have $|f(x^{(k_1)}) - f(x^{(k_2)})| \leq L\|x^{(k_1)} - x^{(k_2)}\|$, and $\|g^{(k)}\| \leq L$.
3. Let $K_\varepsilon = \{k \in \mathbb{N}^* \mid f(x^{(k)}) > v + \varepsilon\}$ be the set of indices such that $x^{(k)}$ is not an ε -optimal solution. Show that $f(x^{(k)}) \rightarrow v$ if and only if K_ε is finite for all $\varepsilon > 0$.
4. Consider $k_1, k_2 \in K_\varepsilon$, such that $k_2 > k_1$. Show that
$$f(x^{(k_1)}) + \langle g^{(k_1)}, x^{(k_2)} - x^{(k_1)} \rangle \leq \underline{v}^{(k_2)} \leq v$$
5. Show that $\varepsilon + f(x^{(k_1)}) + \langle g^{(k_1)}, x^{(k_2)} - x^{(k_1)} \rangle < f(x^{(k_2)})$
6. Show that $\varepsilon < 2L\|x^{(k_2)} - x^{(k_1)}\|$.
7. Prove that $f(x^{(k)}) \rightarrow v$.
8. (Optional - hard) Deduce a complexity bound in terms of the maximum number of points in X that are pairwise at distance at least $\varepsilon/(2L)$.